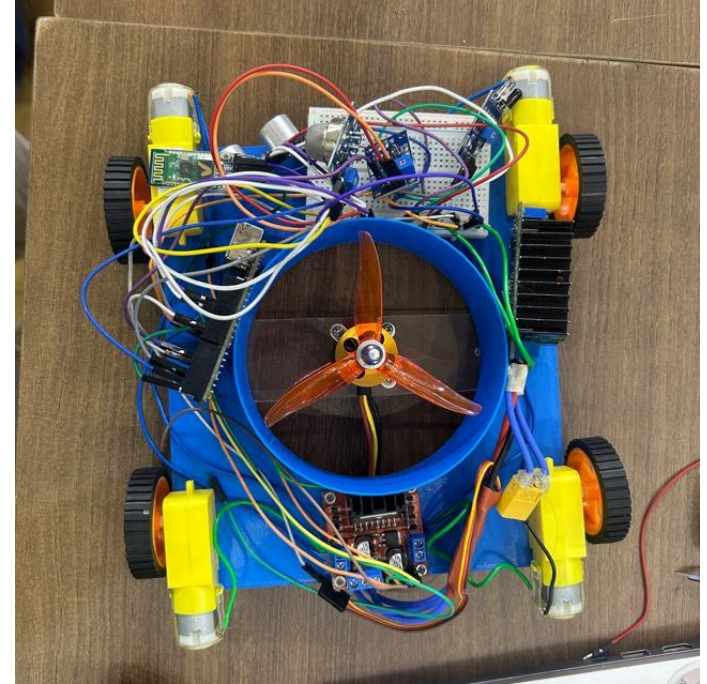


SEEDSTART PITCH DECK

UNCRACKER - A wall climbing robot primarily designed for crack detection

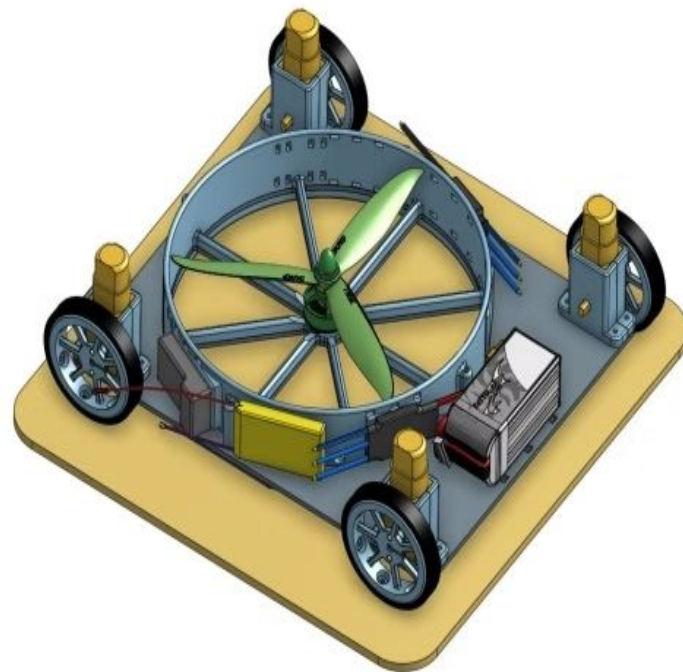
Surya K

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VISION

- **Transform infrastructure inspection:** Replace dangerous, time-consuming manual checks (rope-access, scaffolds) with an autonomous crawler that scales vertical surfaces safely.
- **Innovative combination:** First untethered wall-climber using Electric Ducted Fans (EDF) for adhesion, paired with on-board TinyML analysis and real-time Wi-Fi streaming. This enables continuous, automated defect detection on any wall.
- **High-impact markets:** Target large, fast-growing markets. The global inspection-robotics market is forecast to reach \$9–17 billion by the 2030s. Rapid growth is driven by sectors like oil & gas, power utilities, bridges and buildings, where safety regulations demand better inspection tools.
- **Scalable leader:** Aim to lead a new class of multi-surface inspection robots, scaling to fleets that deliver data-driven maintenance solutions across energy, infrastructure, and manufacturing.



PROBLEM

- **Danger and cost:** Manual wall inspections (e.g. bridge, storage tank, high-rise facade) require people in hazardous positions, using rope-access or cranes. These methods are slow, expensive, and risk human injury.
- **Coverage gaps:** Drones/UAVs cannot inspect close to walls or under overhangs, and are often banned in occupied or confined spaces. Many areas (narrow crevices, behind obstructions) remain unreachable.
- **Limited existing robots:** Current climbers use magnetic tracks (only on steel) or vacuum suction (requiring perfectly flat, smooth surfaces). For example, magnetic crawlers work only on ferrous materials, and suction-based robots lose grip on porous or uneven concrete. No commercial system today uses EDFs. This leaves inspection of concrete towers, painted steel, composites, etc., underserved.
- **Inspection bottleneck:** Defect detection (e.g. cracks, corrosion) is typically manual or post-processed from video, causing delays. In fast-moving industries, real-time anomaly alerts are needed but lacking with current tools.



SOLUTION

- **EDF-based adhesion:** Use an electric ducted fan (EDF) to create a negative-pressure (vortex) underneath the robot, pressing it against the wall. This aerodynamic “suction” provides strong, tunable grip on virtually any surface (concrete, brick, glass, painted metal, composites) without physical seal or magnets. Importantly, no external vacuum pumps or magnets are needed, enabling a fully *untethered* design. Research confirms EDFs can generate adhesion sufficient for wall-climbing robots.
- **On-board TinyML analysis:** Integrate a TinyML-enabled camera system to detect cracks and defects in real time at the edge. The ESP32-CAM module (a \$<10 IoT board) captures video, runs a lightweight neural network on-device, and flags anomalies instantly. Only suspicious frames or alerts are sent upstream, vastly reducing data bandwidth and latency. This on-board inference approach aligns with recent research on TinyML for structural health monitoring (92% accuracy with 40% less energy).
- **Live-stream & control:** The ESP32-CAM also serves as a live video streamer, hosting a web server over Wi-Fi. Inspectors can remotely watch high-definition video and manually verify issues. A simple Bluetooth or Wi-Fi interface (as in [4]) allows direct manual control or override.
- **Modular platform:** The proof-of-concept uses a 4-wheel drive chassis with micro-geared DC motors and an Arduino/ESP32 controller. It carries our EDF (e.g. a 54 mm ducted fan with ~780 g thrust) powered by a lightweight LiPo battery. Future versions will add swappable NDT sensors (ultrasonic, GPR, thermal), but the core innovation is universal adhesion + AI.

SWOT ANALYSIS

Strengths: Novel EDF adhesion allowing climb on varied materials; fully mobile (battery-powered, no tether). On-board TinyML crack detection for immediate alerts. Low-cost hardware (e.g. \$10 ESP32-CAM) and 3D-printed chassis minimize unit cost. Research-backed design (vortex adhesion proven effective).

Weaknesses: Current prototype stage – needs endurance and ruggedness improvements. Prototype battery (1500 mAh) yielded only ~10 min runtime, so flight-time extensions are needed. Payload and on-board compute are limited by size/weight (we must balance adhesion force vs. battery weight). Potential initial reliability unknown (requires extensive field testing).

Opportunities: Inspection robotics market is growing rapidly (~13.9% CAGR) as companies seek safer methods. High-value segments (oil/gas pipelines, power plants, bridges, wind turbines) have severe inspection backlogs. TinyML adoption is accelerating for asset monitoring, enabling new business models (e.g. selling AI-enhanced reports). We can partner with NDT providers (e.g. sensor manufacturers) or service firms to bundle inspections. Future extensions include multi-robot coordination, expanded AI (fuse infrared, LIDAR), and retrofit kits.

Threats: Established players (e.g. Waygate/Baker, Inuktun/Oceaneering, Gecko Robotics) have mature products and customer relationships. Patent/tech risk: others may pursue alternative adhesion patents. Competitive pressure from drone-based inspections (lower-cost UAVs) could emerge. Scaling to real-world reliability is nontrivial (dust, weather, regulatory approvals). Fundraising and time-to-market pressures in a fast-moving tech space.

PROGRESS

Prototype development: We built a lab prototype with a 3D-printed carbon-fiber frame (≈ 82 g), a high-rev ducted fan (BLDC motor, 54 mm ID, 780 g thrust), and four micro-g geared drive wheels (to minimize gap to wall, improving suction). Electronics include an Arduino Nano and an ESP32-CAM board; we added a soft foam skirt to seal small gaps under the fan (as done in [4]) for stable adhesion.

Lab tests: Early trials show the EDF can lift the ~ 200 g robot onto a vertical concrete wall (consistent with [4]). The chassis weight was optimized via cutouts to maximize fan pull. We have confirmed basic motion control (turns, forward/back) and adhesion hold in still-air conditions.

TinyML integration: A convolutional neural network (trained on sample crack/no-crack wall images) is running on the ESP32. In lab tests, the system correctly flagged artificial cracks in the camera view (triggering an alert). We adapted a published TinyML workflow to our hardware; the ESP32's 520 KB RAM and 4 MB PSRAM handle the light model. Frames without anomalies are discarded, minimizing radio use.

Streaming/Demo: The ESP32-CAM hosts a real-time video server (as documented in maker tutorials). We demonstrated live streaming to a browser, with overlay of detection boxes from the TinyML inference.

Next steps: We are iterating the fan shroud and gap height to boost suction (per [3]). Planned milestones include extended battery packs for 30+ min operation, ruggedizing the chassis, and a closed-loop flight test (autonomous wall climbing along a preset path). Customer pilots (e.g. campus buildings) are being arranged for Q3 2025.

BUSINESS ANALYSIS

Market & Segments: The global inspection-robotics market is large and growing. Specifically, oil & gas inspection robotics was \$0.84B in 2025 and is expected to reach \$1.51B by 2034. Major end-markets include oil & gas (tanks, pipes, rigs), power generation (nuclear, thermal plants), water/wastewater (treatment tanks, pipelines), and civil infrastructure (bridges, dams, towers). These sectors face aging assets and strict safety rules.

Customers: Early adopters will be asset owners and service contractors who pay for NDT inspections. Targets include: petrochemical companies (corrosion inspection of vessels), highway agencies (bridge inspections), utilities (nuclear and water), and large building managers (skyscraper façade checks). Our robot can serve as a tool for existing inspection firms (subbed by consultants, specialized NDT contractors).

Value Proposition: Compared to manual or current robotic methods, we offer *faster, safer* inspections. For example, competitors report reducing bridge inspection time from hours to minutes using robots. We provide similar time savings plus capability on non-ferrous structures. Digitally, our AI edge detection means clients get instant defect reports, improving quality of maintenance planning.

Revenue Model: We plan to sell units to inspection companies (price in the low-mid five figures per robot, undercutting heavy industrial systems). Additional services include training, maintenance contracts, and software updates (TinyML model retraining). Over time, we may offer “inspection-as-a-service”: leasing robots and charging per-inspection.

Competition Position: Our EDF-based platform will enter alongside established solutions. We differentiate on capabilities (any-surface climb, built-in AI, portability) and cost. We will build partnerships with NDT sensor makers (e.g. ultrasonic probe integration) to expand offerings. By targeting high-revenue segments first (e.g. petrochemical storage tanks, nuclear containment walls), we aim to capture significant market share in niches where demand is greatest.

INVESTMENT PLAN

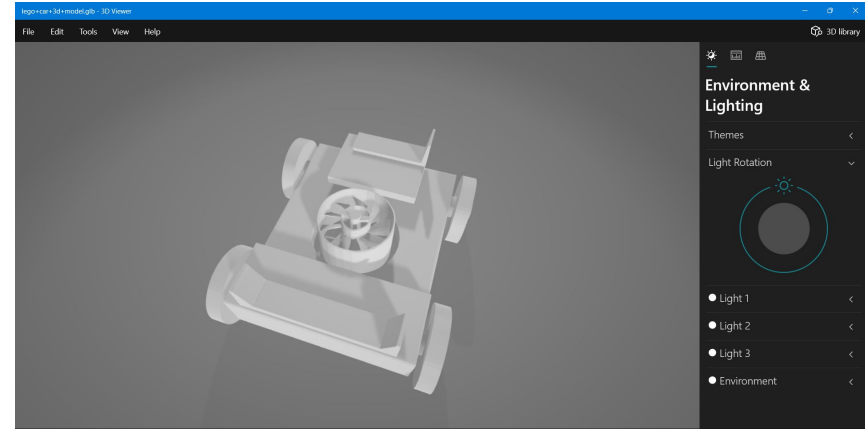
Engineering & Prototyping (~80% of budget): Purchase of components (EDF fans, microcontrollers, sensors), custom PCBs, and rapid iteration tooling. We estimate INR.5000 for hardware builds and design outsourcing.

Equipment: INR.6000 to upgrade lab tools (higher-power battery test bench, oscilloscope, 3D modelling etc.).

Contingency: ~10% of total for unexpected costs.

Cost Efficiency: By using open-source hardware (ESP32, Arduino) and 3D printing, our bill-of-materials per robot is modest (for example, the core camera module is under \$10). We expect cost per unit under \$5K at scale, with significant margin potential. Prototype costs align with industry norms (prototyping often dominates early R&D spend).

Milestones & Use of Funds: Funds will take us through final design and demonstration to target customers. By the end of the seed round, we aim to have: a fully flight-tested robot, at least two pilot inspections booked, and a go-to-market plan for Series A. Our lean plan ensures the investment stretches to key goals.



Custom designed 3D model

COMPETITION

Major competitors include:

- **HausBots (UK)** : Uses patented “AEROGRIP” airflow adhesion (low-pressure vacuum) to climb brick, concrete, metal, even on rough surfaces. Provides heavy-duty tethered robots for bridge and nuclear inspections. **Differentiator:** HausBots is tethered (continuous power) and heavyweight, though it handles uneven surfaces. We match its multi-surface capability but add mobility (no tether) and on-board AI.
- **Waygate/Baker Hughes “BIKE”** : A <10 kg wheeled crawler with *magnetic* tracks for pipe and wall inspection. Lightweight, deployable through manholes, with modular NDT sensors. **Limitations:** Only works on ferromagnetic surfaces (steel pipes, tanks). Our EDF robot, by contrast, can inspect non-ferrous concrete or composites.
- **Gecko Robotics** : (not shown) Magnetic crawlers for tank and heat-exchanger scanning. Focus on data collection; we go beyond by free-climbing ability.
- **Drones (e.g. Flyability Elios)** : Good at general surveys but cannot physically touch or inspect hidden cracks. They also face flight restrictions indoors. Our robot complements drones by sticking to surfaces for close inspection.



Polulu's famous wall climbing robot